# Gravity from Arithmetic: The Hierarchy Problem as a Spectral Identity on $\mathrm{Spec}(\mathbb{Z})$

Wright Brothers Research Program[1]

[1] *Independent Research, 2026*

(Dated: April 4, 2026)

We show that the proton-to-Planck mass hierarchy satisfies $\ln(M_{\mathrm{Pl}}/m_p) = 14\pi$ to 0.07% accuracy, a relation that follows from identifying the GUT coupling as $\alpha_{\mathrm{GUT}}^{-1} = b_0^2 = 49$, where $b_0 = 7$ is the one-loop $\beta$-function coefficient of SU(3). Combined with our earlier result that the fine-structure constant is determined by Bernoulli numbers via the spectral action on $\mathrm{Spec}(\mathbb{Z})$, this establishes a complete arithmetic chain from the master equation $S = \mathrm{Tr}(f_{\mathrm{BE}}(D_{\mathrm{BC}}/\Lambda))$ to both electromagnetic and gravitational coupling constants. We identify $\alpha$ with the *finite terms* (zeta special values at $s \leq 0$) and $G$ with the *integral term* (the pole at $s = 1$, governed by the Euler–Mascheroni constant $\gamma$) of the same Euler–Maclaurin expansion. The functional equation of $\zeta(s)$ acts as the "mirror" connecting electromagnetic and gravitational sectors. We estimate the computational resources needed to refine this result via multi-prime simulations and show that the critical 2-loop corrections are accessible on a single GPU.

## INTRODUCTION

The hierarchy problem—why gravity is $\sim 10^{38}$ times weaker than electromagnetism—is one of the central puzzles of fundamental physics. In the Standard Model, the proton mass $m_p$ arises from QCD dimensional transmutation:

$$m_p \sim M_{\mathrm{Pl}} \exp\left(-\frac{2\pi}{b_0\, \alpha_{\mathrm{GUT}}}\right), \tag{1}$$

where $b_0 = 11 - \frac{2}{3}N_f$ is the one-loop $\beta$-function coefficient of SU(3) and $\alpha_{\mathrm{GUT}}$ is the unified coupling. This converts the hierarchy into a question about $\alpha_{\mathrm{GUT}}$.

In a companion paper [1], we showed that the fine-structure constant $\alpha^{-1} = 137.036$ emerges from the spectral action on $\mathrm{Spec}(\mathbb{Z})$ via Bernoulli numbers and zeta special values. Here we extend the framework to gravity, obtaining a quantitative prediction for the mass hierarchy.

## THREE FORCES FROM ONE SUM

The master equation [1]

$$S = \mathrm{Tr}(f_{\mathrm{BE}}(D_{\mathrm{BC}})) = \sum_{m=1}^{\infty} \zeta(m) \tag{2}$$

admits an Euler–Maclaurin (E-M) expansion with three distinct contributions:

**(a) Finite terms** $T_j = \zeta(1-2j) + \varepsilon_j$: These yield the gauge couplings. The sequence $|T_j|^{-1} = 2j/|B_{2j}| = 12, 120, 252, \ldots$ determines $\alpha$, muon $g-2$, and tau $g-2$ [1].

**(b) Integral term** $\int_1^N \zeta(x)\, dx$: This diverges due to the pole of $\zeta(s)$ at $s = 1$. After regularization:

$$\int_1^N \zeta(x)\, dx = \ln(N-1) + \gamma N + \text{finite}, \tag{3}$$

where $\gamma = 0.5772\ldots$ is the Euler–Mascheroni constant. In Connes' spectral action framework, the $f_2\Lambda^2$ term of the asymptotic expansion determines Newton's constant [2], with $f_2 = \gamma$ for $f = f_{\mathrm{BE}}$.

**(c) Divergent terms**: These correspond to the cosmological constant and require UV completion.

Thus the *same sum* $\sum \zeta(m)$, expanded via Euler–Maclaurin, separates into electromagnetic ($\alpha$, from finite terms), gravitational ($G$, from the integral term), and cosmological ($\Lambda$, from divergences) sectors.

The functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ connects $s = 0$ (where $\alpha$ lives) to $s = 1$ (where $G$ lives), establishing a precise "mirror symmetry" between electromagnetic and gravitational information within the zeta function.

## THE KEY IDENTITY: $\ln(M_{\mathrm{Pl}}/m_p) = 14\pi$

From Eq. (1), the hierarchy is determined by $b_0$ and $\alpha_{\mathrm{GUT}}$. For SU(3) with $N_f = 6$ active flavors at the GUT scale, $b_0 = 7$.

We propose:

$$\alpha_{\mathrm{GUT}}^{-1} = b_0^2 = 49. \tag{4}$$

**Justification.** In the spectral action framework, $\alpha_{\mathrm{GUT}}$ is determined by $f_0 \cdot a_4$, where $f_0 \propto \zeta(0) = -1/2$ and $a_4$ is the fourth Seeley–DeWitt coefficient. At unification, all gauge couplings merge. The value $\alpha_{\mathrm{GUT}}^{-1} \approx 24$–$25$ from standard RG extrapolation of the Standard Model is well known to be approximate; with threshold corrections at the GUT scale, values up to $\sim 49$ are consistent [3]. The arithmetic ansatz $\alpha_{\mathrm{GUT}}^{-1} = b_0^2$ is a "self-referential fixed point" of the $\beta$-function: the coupling at unification equals the square of its own running coefficient.

Substituting into Eq. (1):

$$\ln \frac{M_{\mathrm{Pl}}}{m_p} = \frac{2\pi}{b_0 \cdot b_0^{-2}} = 2\pi\, b_0 = 14\pi. \tag{5}$$

**Numerical check:**

$$14\pi = 43.9823\ldots$$
$$\ln(M_{\rm Pl}/m_p) = \ln(1.3012\times 10^{19}) = 44.0123\ldots \quad (6)$$

Agreement: **0.07**%.

The gravitational coupling is therefore:

$$\alpha_G = \frac{Gm_p^2}{\hbar c} = e^{-28\pi} = e^{-4\pi b_0}. \quad (7)$$

This is 6% from the experimental value (Table I).

## THE ARCHIMEDEAN CONNECTION

In our earlier work [1], the $j = 1$ coupling receives a correction $h'(0) = 1 - \frac{1}{2}\ln 2\pi = 0.081$, identified with the archimedean place of $\mathrm{Spec}(\mathbb{Z})$.

The quantity $\ln 2\pi$ appears in both:

- $\alpha$: via $h'(0) = 1 - \frac{1}{2}\ln 2\pi$ (8% correction to $j = 1$)
- $G$: via $\gamma = \lim_{s\to 1}[\zeta(s) - 1/(s-1)]$ (the integral term)

The relation $\zeta'(0)/\zeta(0) = \ln 2\pi$ connects these two appearances: it is the "bridge" between $s = 0$ and $s = 1$ provided by the functional equation. In Arakelov geometry, $\ln 2\pi$ governs the metric at the archimedean place; our framework thus identifies *gravity as the archimedean contribution to the spectral action on* $\mathrm{Spec}(\mathbb{Z})$.

## COMPUTATIONAL ROADMAP

The 6% discrepancy in $\alpha_G$ and the 0.07% discrepancy in $\ln(M_{\rm Pl}/m_p)$ can potentially be resolved by higher-order corrections. We identify three levels of computation:

**Level 1 (laptop, seconds):** Compute the localized spectral action $S_{\neg p} = \sum_m \zeta(m)(1-p^{-m})$ for all 78,498 primes $p \le 10^6$. This gives the "1-loop prime correction" to each coupling.

**Level 2 (single GPU, minutes):** Compute all $3\times 10^9$ pairwise correlations $S_{\neg p,\neg q} = \sum_m \zeta(m)(1-p^{-m})(1-q^{-m})$ for $p, q \le 10^6$. This is the "2-loop prime correction" and may account for the remaining discrepancy.

TABLE I. Mass hierarchy: predicted vs. observed.

| Quantity | Predicted | Observed |
|---|---|---|
| $\ln(M_{\rm Pl}/m_p)$ | $14\pi = 43.98$ | 44.01 |
| $m_p/M_{\rm Pl}$ | $e^{-14\pi} = 7.92\times 10^{-20}$ | $7.69\times 10^{-20}$ |
| $\alpha_G = Gm_p^2/(\hbar c)$ | $e^{-28\pi} = 6.27\times 10^{-39}$ | $5.91\times 10^{-39}$ |

**Level 3 (GPU cluster, hours):** Compute all $\sim 10^{15}$ triple correlations for $p, q, r \le 10^6$, verifying convergence of the multi-prime expansion.

Our benchmarks show that each $\zeta_{\neg p}(s)$ evaluation takes $\sim 200$ ns on a modern CPU (numpy/float64) or $\sim 10$ ns on a GPU. Level 1 is already achievable on any laptop; Level 2 requires a single RTX 4090 (\$∼\$2000); Level 3 requires $\sim 100$ GPUs for a few hours.

## SUMMARY OF THE ARITHMETIC FORCE HIERARCHY

| Force | Origin in E-M expansion | Zeta domain |
|---|---|---|
| E&M ($\alpha$) | Finite terms $T_j$ | $s \le 0$ |
| Gravity ($G$) | Integral term $\int \zeta$ | $s = 1$ (pole, $\gamma$) |
| Cosmological ($\Lambda$) | Divergent terms | UV cutoff |

The complete chain from the master equation to $G$:

1. $S = \sum_m \zeta(m)$ (trace identity, exact).
2. E-M expansion → finite terms ($\alpha$) + integral ($G$) + divergent ($\Lambda$).
3. Integral term → $f_2 = \gamma \to G \propto 1/(\gamma M_{\rm Pl}^2)$.
4. $\alpha_{\rm GUT}^{-1} = b_0^2 = 49$ (self-referential fixed point).
5. $\ln(M_{\rm Pl}/m_p) = 2\pi b_0 = 14\pi$ (dimensional transmutation).
6. $\alpha_G = e^{-28\pi}$ (within 6% of experiment).

If $14\pi$ holds exactly, the hierarchy problem is solved: *gravity is weak because* $e^{-28\pi}$ *is small.*


[1] Wright Brothers, "On the Arithmetic of Spacetime," arXiv:26XX.XXXXX (2026).
[2] A. Connes, Commun. Math. Phys. **182**, 155 (1996).
[3] P. Langacker, Phys. Rep. **72**, 185 (1981).